Experiment 9: Lid-Driven Cavity Flow (SIMPLE)

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1 Aim

Solve the 2D incompressible lid-driven cavity at $Re = 100$ using the SIMPLE algorithm on a collocated grid, and validate the centreline velocity against the benchmark data of Ghia et al. (1982).

2 Assumptions & Limitations

- **Incompressible, Laminar, 2D:** Constant density and viscosity; no turbulence model.
- **Pseudo-transient Marching:** The steady solution is reached by marching in time with $\Delta t = Co\,\Delta x/u_{lid}$ until the mass residual is below tolerance.
- **Collocated Grid:** u, v, p are all stored at the same nodes, without Rhie–Chow interpolation.
- **Single Sweep for p':** One Jacobi-type update of the pressure correction is made per SIMPLE iteration (the inner loop is commented out in the code).
- **Laminar Limit:** The solver cannot represent turbulence; the Ghia data for $Re = 400, 1000, 10000$ are stored for flexibility, but only $Re = 100$ is a valid laminar case here.

3 Governing Equations & Simplification

For constant ρ and ν, the steady incompressible Navier–Stokes equations are

$$\nabla \cdot \mathbf{u} = 0, \qquad (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} \tag{1}$$

In 2D components:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) \tag{2}$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial y} + \nu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) \tag{3}$$

The time derivative is kept only as a *relaxation device* (pseudo-time) towards steady state. The Reynolds number fixes the viscosity:

$$Re = \frac{U_{lid} L}{\nu} = 100 \;\Rightarrow\; \nu = \frac{U_{lid} L}{Re} = 0.01 \tag{4}$$

4 Discretization & Algorithm

4.1 Finite-Volume Momentum Equations

Integrating over a control volume with implicit Euler time stepping:

$$a_P \phi_P^* = \sum a_{nb} \phi_{nb}^* + a_P^0 \phi_P^n + (\text{pressure term}), \qquad a_P = \sum a_{nb} + a_P^0, \quad a_P^0 = \frac{\rho \Delta x \Delta y}{\Delta t} \tag{5}$$

with face flux $F_f = \rho u_f A_f$, diffusion conductance $D_f = \nu A_f / \Delta$ and cell Péclet number $\mathrm{Pe}_f = F_f / D_f$. Neighbour coefficients follow the **Power-Law scheme**:

$$a_E = D_e A(|\mathrm{Pe}_e|) + \max(-F_e, 0), \quad a_W = D_w A(|\mathrm{Pe}_w|) + \max(F_w, 0), \quad A(\mathrm{Pe}) = \max[0, (1 - 0.1|\mathrm{Pe}|)^5] \tag{6}$$

(a_N, a_S analogous). Face velocities are the arithmetic means of adjacent nodes, and the pressure gradient is a central difference over two cells, $(p_W - p_E)\Delta y/2$ and $(p_S - p_N)\Delta x/2$.

4.2 Pressure Correction

With $u = u^* + u'$, $p = p^* + p'$ and the SIMPLE approximation (neighbour corrections neglected):

$$u'_P = d_e \frac{p'_W - p'_E}{2}, \quad v'_P = d_n \frac{p'_S - p'_N}{2}, \qquad d_e = \frac{\Delta y}{a_P}, \; d_n = \frac{\Delta x}{a_P} \tag{7}$$

Substituting into continuity gives

$$a'_P p'_P = a'_E p'_E + a'_W p'_W + a'_N p'_N + a'_S p'_S + b, \quad b = \rho \frac{u^*_W - u^*_E}{2} \Delta y + \rho \frac{v^*_S - v^*_N}{2} \Delta x \tag{8}$$

where b is the mass imbalance of the guessed field and is also used as the convergence monitor, $\max |b| < 10^{-6}$.

4.3 Update and Under-relaxation

$$p = p^* + \alpha_p p', \qquad u = u^* + u', \quad v = v^* + v', \qquad \alpha_p = 0.3 \tag{9}$$

Pressure under-relaxation prevents divergence caused by the SIMPLE neglect of neighbour corrections.

4.4 Algorithm

1. Initialize $u = v = p = 0$ with the lid velocity $u = U_{lid}$ on the top row.
2. Solve the momentum equations for u^*, v^* using the current p^*; enforce wall BCs.
3. Solve the pressure-correction equation; update p (relaxed) with zero-gradient walls.
4. Correct velocities; re-apply BCs.
5. Check $\max|b| < 10^{-6}$; otherwise repeat (up to $n_t = 20000$ iterations).

5 Code Structure

- `all u, v, p stored on the same (ny, nx) mesh` – collocated arrangement.
- `Power Law Scheme` – function `A(P)` for convection–diffusion weighting.
- `u_old, v_old: previous time level | u, v, p: current guess` – ϕ^n for the transient term; ϕ^* for the SIMPLE iterate.
- `faces (uniform grid)` – face values by simple averaging.
- `/2: central difference over 2 cells` – pressure-gradient and mass-flux terms span $i-1$ to $i+1$.
- `internal loop for time relaxation` (commented out) – a single sweep of the p' equation is done per iteration; inner sweeps could be enabled for a tighter p' solution.
- `zero gradient at walls` – Neumann condition for p and p'.
- `p = p* + alpha_p p'` and `u = u* + u', v = v* + v'` – the SIMPLE update step.
- `Validation with Ghia et al.` – tabulated v at $y = 0.5$ for $Re = 100, 400, 1000, 10000$ at the Ghia abscissae `x_test`; only the $Re = 100$ set is plotted.

6 Methodology and Boundary Conditions

Parameter	**Value**	**Parameter**	**Value**
Cavity size ($L_x = L_y$)	1.0	Grid ($n_x \times n_y$)	60×60
Lid velocity U_{lid}	1.0	Density ρ	1.0
Reynolds number	100	Kinematic viscosity ν	0.01
Courant number Co	0.5	Time step $\Delta t = Co\,\Delta x/U_{lid}$	$\approx 8.5 \times 10^{-3}$
Pressure relaxation α_p	0.3	Tolerance	1.0×10^{-6}

Table 1: Parameter set for the lid-driven cavity.

- **Top wall (lid):** $u = U_{lid} = 1$, $v = 0$.
- **Left, right and bottom walls:** no-slip, $u = v = 0$.
- **Pressure and p':** zero normal gradient on all walls (pressure level is arbitrary in a closed cavity).

7 Results & Discussions

Figure 1 shows the velocity-magnitude contours and streamlines. The lid drives a primary vortex whose centre lies slightly above the cavity centre and shifted towards the lid-moving side, as expected at $Re = 100$, with the strongest velocities near the lid.

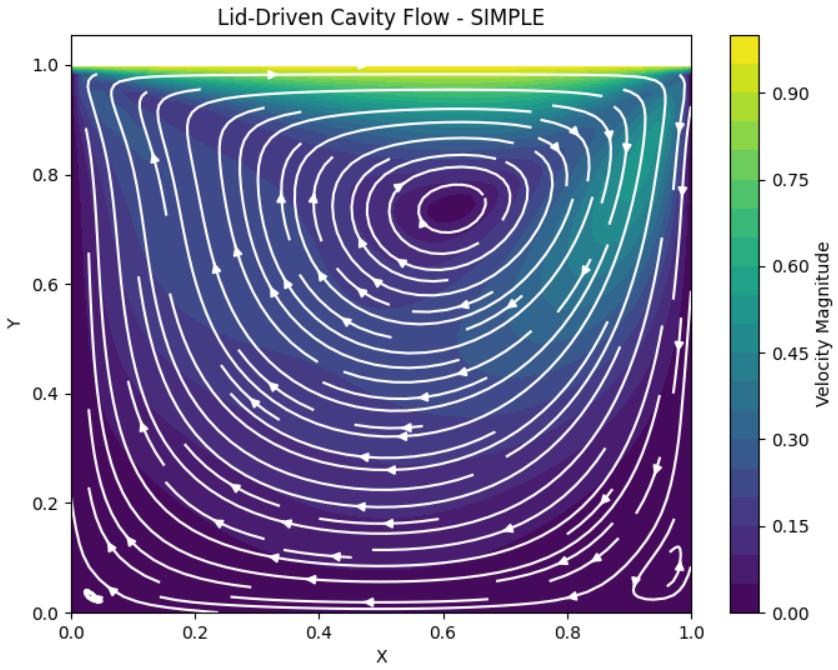


Figure 1: Velocity magnitude and streamlines for the $Re = 100$ lid-driven cavity.

Validation: Figure 2 compares the computed v-velocity along $y = 0.5$ with the benchmark of Ghia et al. The numerical curve passes almost exactly through the benchmark points, validating the SIMPLE implementation, the Power-Law discretization and the boundary treatment.

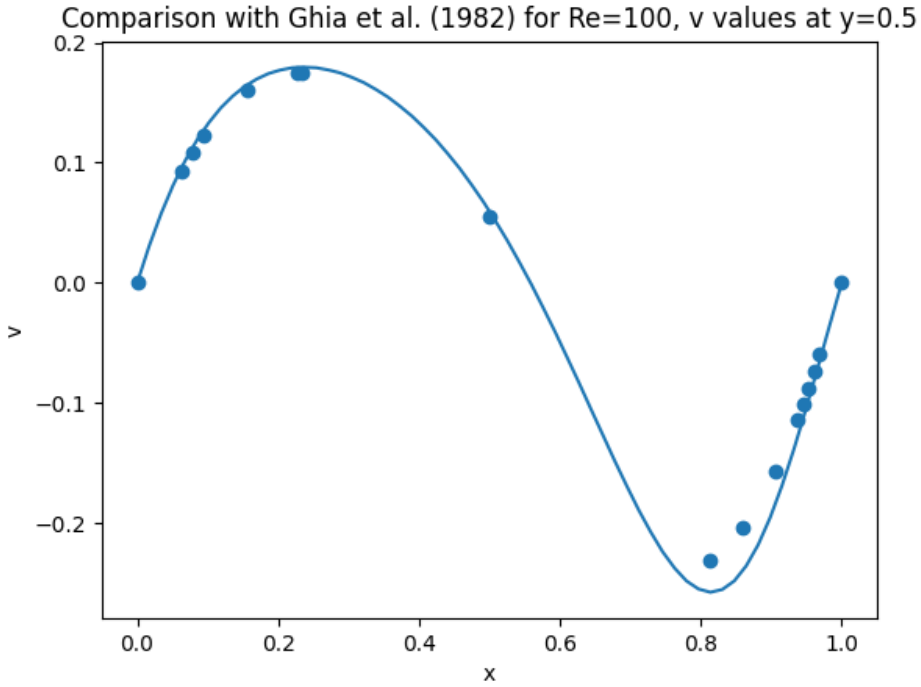


Figure 2: Comparison of $v(x)$ at $y = 0.5$ with Ghia et al. (1982) at $Re = 100$.

Higher Reynolds numbers: The code carries benchmark data for $Re = 400$, 1000 and 10000, and Re can be changed freely. However, the very high-Re flows are unsteady/turbulent, and a laminar steady-state model is not valid there; a turbulence closure (as in Experiment 10) would be needed.

8 Conclusion

The SIMPLE algorithm on a collocated grid with Power-Law convection–diffusion discretization and pressure under-relaxation reproduces the benchmark lid-driven cavity solution at $Re = 100$. The method is validated against Ghia et al. with , and its limits (laminar regime, no Rhie–Chow correction, single p' sweep) are identified for future extensions.